



**Faculty of Computers &  
Artificial Intelligence**



**Benha University**

**E-commerce project  
for selling electronic products using AI**

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# E-commerce project

(website and mobile application) to sell electronic products using AI



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# CHAPTER ONE

# Introduction

The Online Shopping System is a modern e-commerce platform designed to provide users with a seamless, efficient, and secure way to purchase products online. In today's digital age, where convenience and timesaving are critical, this system aims to simplify the shopping process by offering a unified platform for both **web** and **Android** applications. The platform is tailored for electronics products, enabling users to browse, select, and purchase items such as smartphones, laptops, and accessories with ease.

The system features a user-friendly interface built using **React** for the **web frontend**, ensuring a responsive and dynamic user experience. It also includes robust search and filtering capabilities, secure payment processing, and efficient order management. The platform caters to both customers and administrators, offering functionalities such as user registration, shopping cart management, order tracking, and product catalog management. By digitizing the shopping experience, the Online Shopping System eliminates the need for physical store visits, saving time and effort for users while providing businesses with a scalable and efficient solution.

One of the key enhancements of this system is the **AI-powered product recommendation engine**. This feature analyzes user behavior and preferences, offering personalized product suggestions to enhance the shopping experience. By leveraging machine learning algorithms, the system adapts to the user's browsing history and purchasing patterns, ensuring that the product recommendations are highly relevant and increase the likelihood of conversions.

This project addresses the growing demand for online shopping by creating a reliable, scalable, and secure e-commerce platform. It leverages advanced technologies, including **React** for the frontend, **.NET** for the **backend**, **Kotlin** for the Android app, and **AI** for personalized recommendations, to deliver a high-performance system that meets the needs of modern consumers and businesses.

## Problem definition

The current e-commerce landscape lacks a **unified platform** that delivers a **consistent and seamless user experience** across both **web and mobile applications**.

Many existing platforms either do not offer a dedicated mobile app or suffer from poorly designed user interfaces, making it challenging for users to browse and purchase electronics efficiently.

This issue is particularly pronounced in the **electronics category**, where users often require detailed product information, advanced search and filtering options, and a smooth purchasing process.

## What is the importance of this problem

1. **High Demand for Electronics:** Electronics are among the most purchased products online, and users expect a specialized shopping experience tailored to their needs.
2. **Cross-Platform Shopping:** Consumers increasingly switch between devices (web and mobile) while shopping, and a lack of consistency across platforms can lead to frustration and abandoned purchases.
3. **User Experience Matters:** A poorly designed interface or inefficient search functionality can drive users to competitors, resulting in lost sales and reduced customer loyalty.
4. **Security Concerns:** With the rise in online transactions, users demand secure platforms to protect their personal and payment information.

Addressing these issues is critical to meeting user expectations, improving customer satisfaction, and staying competitive in the e-commerce industry.

## What are the current solutions

Existing e-commerce platforms, such as Amazon, eBay, and others, provide general-purpose solutions for online shopping. However, they have several limitations:

### 1. Lack of Specialization:

- Most platforms are not tailored for electronics, missing features like detailed product specifications, compatibility checks, or advanced filtering options.

### 2. Inconsistent Cross-Platform Experience:

- Many platforms do not offer seamless experience across web and mobile apps, leading to a fragmented user journey.

### 3. Complex User Interfaces:

- Some platforms have cluttered or outdated designs, making it difficult for users to navigate and complete purchases.

### 4. Limited Security Features:

- Not all platforms prioritize secure payment processing and data protection, which can deter users from completing transactions.

## How Will Your Solution Solve the Problem?

### What is new?

Our solution is an e-commerce platform specifically designed for selling electronics, with support for both web (using React) and Android (using Kotlin) platforms.

Here's how it solves the problem and introduces innovative features:

### How It Solves the Problem

#### 1. Unified Cross-Platform Experience:

- The platform provides a consistent and seamless user experience across web and Android applications, ensuring users can switch between devices without disruption.

#### 2. User-Friendly Interface:

- A modern, intuitive design makes it easy for users to browse, search, and purchase electronics without unnecessary complexity.

#### 3. Advanced Search and Filtering:

- Tailored for electronics, the platform offers advanced filtering options (e.g., by brand, price range, specifications) to help users find products quickly.

#### 4. Detailed Product Information:

- Comprehensive product descriptions, high-quality images, and customer reviews enable users to make informed purchasing decisions.

#### 5. Secure Transactions:

- Integration with secure payment gateways (e.g., Stripe, PayPal) and robust authentication mechanisms ensures the safety of user data.

# What is New?

## 1. Focus on Electronics:

- Unlike generic e-commerce platforms, our solution is specifically designed for electronics, offering features like compatibility checks and technical specifications.

## 2. Cross-Platform Consistency:

- The platform provides a unified experience across **web (React)** and **Android (Kotlin)** apps, addressing the fragmentation seen in many existing solutions.

## 3. Enhanced User Experience:

- The use of **React** for the web frontend ensures a responsive and dynamic interface, while **Kotlin** for the Android app delivers a smooth mobile experience.

## 4. Scalable and Secure Backend:

- The backend, built using **.NET**, ensures high performance, scalability, and security, making the platform suitable for growing businesses.

## 5. Specialized Features for Electronics:

- The platform includes tools like **product comparison, compatibility checks, and technical specifications**, which are essential for electronics shoppers.

By addressing these gaps and introducing innovative features, our solution provides a **specialized, efficient, and secure shopping experience** for electronics enthusiasts, setting it apart from generic e-commerce platforms.

# Scope

The scope of this project is to develop a **cross-platform e-commerce platform** specifically designed for selling **electronics products**, with support for both **web and Android applications**.

The platform will provide seamless and consistent user experience across devices, enabling users to browse, select, and purchase electronics with ease.

## Key Components of the Project

### 1. Android Application:

- Developed using **Kotlin**, the Android app will provide a smooth and intuitive mobile shopping experience.
- Features include user registration, product browsing, shopping cart management, and secure payment processing.

### 2. Web Application:

- Built using **React**, the web app will offer a responsive and dynamic interface for users to shop from their browsers.
- It will mirror the functionality of the Android app, ensuring consistent experience across platforms.

### 3. AI User Assistance:

- Python for writing the code
- Colab environment for training models
- Kaggle to get datasets

## 4. Data Analysis

- **Data Collection & Cleaning** – Gather and preprocess e-commerce data, ensuring accuracy and consistency for analysis.
- **Sales & Customer Insights** – Analyze sales trends, customer behavior, and website traffic to provide actionable insights for business decisions.
- **Performance Tracking** – Develop dashboards and reports using tools like Power BI or Tableau to monitor key performance indicators (KPIs).
- **Predictive Analytics** – Use statistical models and machine learning techniques to forecast sales, optimize pricing, and improve user experience.

## 5. Backend Development:

- .NET for the Backend
- ASP.NET Web APIs (MC)
- Web Service - Web APIs
- JWT (JSON Web Token) – Authentication & security
- RESTful web services

### 1.Design Patterns for ASP.NET Web APIs:

#### a) Repository Pattern

Purpose: Separates data access logic from the business logic.

Benefits: Improves maintainability, testability, and reduces code duplication.

#### b) Unit of Work Pattern

Purpose: Manages database transactions efficiently, ensuring data consistency.

Usage: Typically used with the Repository Pattern.

### 2. Architecture Patterns for ASP.NET Web APIs:

#### Layered Architecture (N-Tier) Structure:

- i.Presentation Layer (Controllers)
- ii.Business Logic Layer (Service Layer)
- iii.Data Access Layer (Repositories/EF Core)
- iv.Database Layer

## 6. Database Management:

- A robust database will be designed to store and manage data related to users, products, orders, and payments.
- The database will ensure efficient data retrieval and storage, supporting the platform's scalability.

## Key Features

- **24/7 Accessibility:**
  - The platform will provide **24×7 service**, allowing customers to browse the website, place orders, and track deliveries at any time.
- **User Registration and Authentication:**
  - Users can create accounts, log in, and manage their profiles securely.
- **Product Browsing and Search:**
  - Users can browse and search for electronics products using advanced filters (e.g., brand, price range, specifications).
- **Shopping Cart and Checkout:**
  - Users can add products to their cart, review their selections, and proceed to checkout.
- **Payment Processing:**
  - Secure payment gateways (e.g., Stripe, PayPal) will be integrated to ensure safe and reliable transactions.
- **Admin Panel:**
  - Administrators can manage products, update inventory, and view sales reports.

## Future Enhancements

The platform is designed with scalability in mind, allowing for future modifications and additions. Some potential improvements include:

### 1. Integration with Local Shopkeepers:

- Local shops can use the platform to provide an online interface for customers, enabling them to shop and place orders directly.
- This will benefit customers by saving time and supporting local businesses, as people can shop from nearby stores instead of relying on large supermarkets.

### 2. Efficient Delivery System:

- Deliveries from local vendors will be faster, with products delivered on the same day or the next day, depending on availability.
- Customers will be informed about delivery timelines, allowing them to confirm or modify their orders accordingly.

### 3. Easy Returns and Exchanges:

- Returns and exchanges will be simplified, as delivery personnel can handle them directly due to the proximity of local stores.

### 4. Chatbot for Customer Support:

- A chatbot can be integrated to provide instant support, allowing customers to resolve queries or issues in real-time.

## Product Catalog

The platform will focus exclusively on **electronics products**, including:

- **Smartphones**
- **Laptops and Computers**
- **Accessories** (e.g., headphones, chargers, smartwatches)
- **Home Electronics** (e.g., smart home devices, cameras)

## Limitations

- The platform will initially focus on electronics and will not include other product categories.
- The scope is limited to **web and Android platforms**; iOS support may be considered in future updates.

By defining a clear scope, this project ensures a focused and efficient development process, delivering a **specialized, user-friendly, and secure e-commerce platform** for electronics.

# CHAPTER TWO

# Introduction

System analysis is the process of **gathering and interpreting facts, diagnosing problems**, and using this information to recommend improvements to a system. It is a **problem-solving activity** that requires intensive communication between **system users** and **system developers** to ensure the final solution meets user needs and expectations.

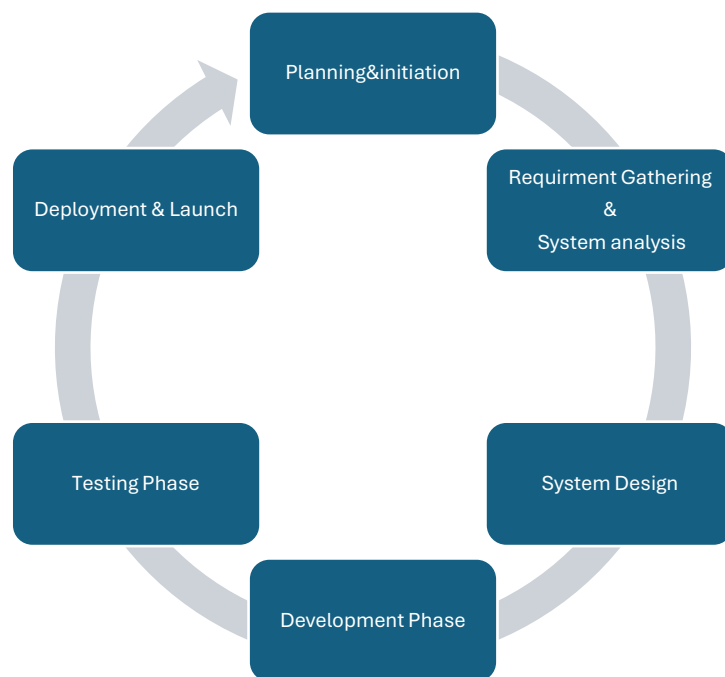
System analysis is a critical phase in the system development process. During this phase, the system is viewed as a **whole**, and its **inputs, processes**, and **outputs** are closely examined to identify problem areas. Based on this analysis, potential solutions are proposed and reviewed with users. Feedback is incorporated, and the proposal is refined until the user is satisfied. This iterative process ensures that the final system aligns with user requirements and business goals.

The goal of system analysis is to create a **blueprint** for the system that addresses current challenges, improves efficiency, and provides a foundation for future enhancements. This phase lays the groundwork for the design and implementation of a system that is **user-friendly, scalable**, and **aligned with organizational objectives**.

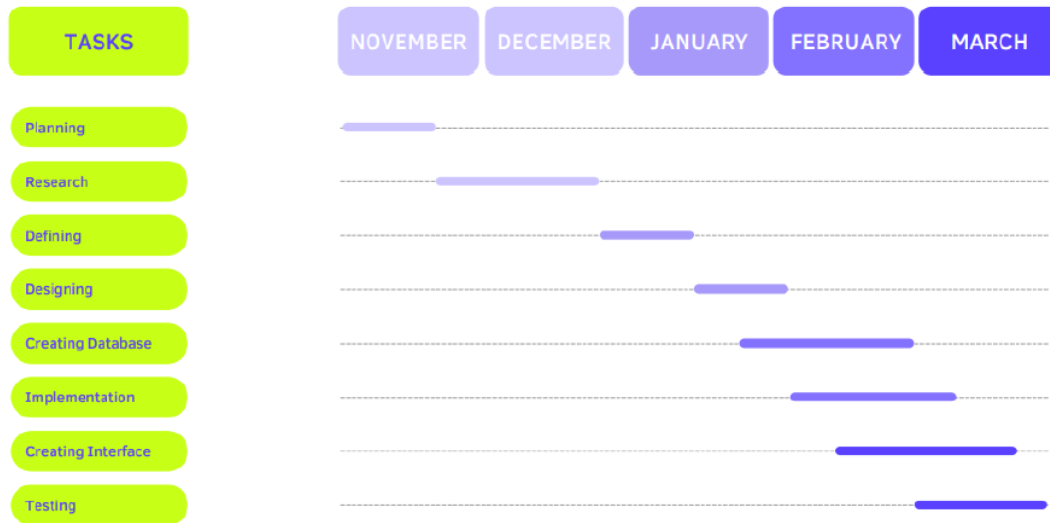
# Planning

## System Development Life Cycle (SDLC)

Software Development Life Cycle (SDLC) is a structured approach to software development that outlines the stages and processes involved in building, deploying, and maintaining software applications. The SDLC serves as a guideline for creating high-quality software in an efficient and predictable manner, ensuring that projects are completed on time, within budget, and meet specified requirements.



## Gantt Chart



Gantt chart displaying a project timeline with various phases and tasks. It outlines key stages such as Project Planning & Initiation, Requirements Gathering, System Design, Development, Testing, Deployment, and Post-Launch Support. Each phase has a specific duration and assigns tasks, with dependencies between them. The timeline spans from November 2024 to ongoing post-launch support, with critical tasks highlighted in yellow. The orange bars represent the scheduled execution periods for each task.

## User and System Requirements

To ensure the successful development and deployment of the e-commerce platform, the following **user and system requirements** have been identified.

These requirements are categorized into **functional requirements** (what the system should do) and **non-functional requirements** (how the system should perform).

## Functional Requirements

### User Authentication:

- Users must be able to create accounts.
- The system should authenticate users securely, requiring a username and password.

### User Profiles:

- Each user should have a profile with personal information, such as name, age, and address.

### Search bar:

- Automatic search suggestions: Once the user starts typing, suggestions for products or categories related to the entered word should appear.
- If the products are not available, suggest other links to the location of the product.

### Product specifications:

- Technical characteristics: such as size, weight, available colors, details of any special techniques, and product photos.
- Ratings and reviews  
Add reviews from previous customers if available, as this helps build trust.

### Add products to the cart:

- Add to cart: Allows the user to add products to the shopping cart with the ability to modify the quantity or delete products.
- Retain cart: The application should retain the cart even if the user logs out or closes the application.

### Personal account management:

- Modifying personal data: enabling the user to modify his personal information such as name, address, email, etc.
- View order history: View previous order history with details about each purchase (date, products, status).

# Non-Functional Requirements

## Performance:

- The application's response must be fast and efficient, especially at checkout moments or when adding products to the cart.
- The application must handle many users at the same time without affecting performance.

## Scalability:

- The application must be able to scale to accommodate growth in the number of products, customers, and sales.
- It should be able to easily add new functionality, such as adding new payment methods or expanding shipping options.

## Security:

- Data Protection: Sensitive data such as user data (payment information, email, address) must be secured using advanced encryption techniques.
- Identity Verification: The app should be able to verify users' identity via multiple methods such as email or text messages.

## Reliability:

- The application must always be reliable and reliable.
- It must have mechanisms to automatically restart or switch over in the event of a technical error.

## Usability:

- The user interface should be intuitive and accessible to users of different age groups.
- It should be easy for users to find products, add items to cart, and complete the purchase.

## Compatibility:

- The application should be compatible with the latest versions of major web browsers.
- It should be compatible with different internet browsers (Chrome, Firefox, Safari, etc.).

**Data Backup:**

- The system should perform regular backups of user data and application databases.
- Data recovery mechanisms should be in place in case of unforeseen issues.

These requirements provide a balance of functional features and non-functional characteristics that are crucial for the success, performance, and user satisfaction of your store application. Adjustments can be made based on specific project goals and constraints.

# FEASIBILITY STUDY

## Technical Study:

The system is an application on which requires:

### Frontend (Web Application)

**Framework:** **React.js**: A popular JavaScript library for building responsive and dynamic user interfaces.

**Styling:** **CSS3** or **SASS**: For styling the web application.

**Bootstrap or Material-UI**: For pre-built UI components and responsive design.

**State Management:** Redux or Context **API**: For managing global state in the application.

### Frontend (Android Application)

**Programming Language:** **Kotlin** The preferred language for Android development.

**UI Framework:** **XML** Layouts: Traditional way to design Android UIs.

**State Management:** View Model and Live Data: For managing UI-related data.

**Networking:** **Retrofit** For making **API** calls to the backend.

## Backend

**Framework:** **.NET** (ASP.NET Core) A robust framework for building scalable and secure backend systems.

**API Development:** **RESTful APIs** for communication between the frontend and backend.

**Authentication:** **JWT** (JSON Web Tokens) For secure user authentication.

**Payment Gateway Integration:**

**Stripe** or **PayPal**: For secure payment processing.

## AI Integration

**Recommendation Engine:** The system integrates an **AI-powered** recommendation engine to enhance the shopping experience. It uses machine learning algorithms to analyze user behavior, such as browsing history and past purchases, to provide personalized product suggestions.

**AI Framework:** TensorFlow or Scikit-learn – For building and deploying machine learning models that power the recommendation system.

**Data Analysis:** The system collects data from user interactions and applies predictive models to suggest products users are most likely to be interested in, increasing engagement and conversion rates.

Each of those technologies are freely available and technical skills are manageable.

## Financial Study

The system will follow freeware software standards, ensuring that no costs will be charged to potential customers. By using open-source tools and freely available resources, the financial burden on users is minimized, making the platform accessible to a wider audience. The focus is on providing high-quality user experience without imposing financial barriers.

## Resource Feasibility

The resources required to successfully implement and maintain the system include:

- **Programming Device:** Standard computing devices with sufficient processing power will be needed for development. This includes laptops or desktops with modern specifications capable of running the required development environments for **React.js**, **Kotlin**, **.NET**, and **AI** frameworks.
- **Hosting Space:** Cloud hosting services (such as AWS, Google Cloud, or Azure) will be used for deploying the web and backend services. Free-tier services or cost-efficient options will be explored for initial development and scaling.
- **Programming Tools:** Free and open-source tools will be used for development:
  - React.js for frontend development
  - Kotlin and Android Studio for the Android application
  - .NET (ASP.NET Core) for backend development
  - AI libraries like TensorFlow or Scikit-learn for recommendation engines
  - GitHub for version control and collaboration

- **Programming Individuals:** The team consists of developers skilled in React, Kotlin, .NET, and machine learning technologies. As this project utilizes free tools and resources, the need for external licenses is minimized, and the development effort primarily involves internal expertise.

Thus, the project is resource-feasible, relying on available free tools and technologies to minimize costs while ensuring that all required components are included.

## Risk Feasibility

The risks associated with the size of the database created or used by the system are manageable. The data handled by the platform, such as user profiles, product listings, and purchase transactions, is within the scale that can be supported by Python-based tools and scalable database solutions (such as PostgreSQL or MongoDB).

As the system is designed to handle typical e-commerce data, the risk of exceeding capacity is minimal, and performance can be maintained with proper optimization and scaling techniques. Regular backups and cloud-based solutions will be used to ensure data integrity and reliability

.

## Social Feasibility

The system uses freely available tools and is intended to be an open-source solution. By adopting open-source standards, the project ensures wider access to technology, allowing anyone to contribute or use the system without financial constraints. This approach promotes collaboration and knowledge sharing within the developer community, benefiting both users and businesses.

Furthermore, by offering an accessible e-commerce platform, the system supports the growing demand for online shopping in the Middle East, making it an inclusive solution that provides convenience to users without additional costs.

# Stakeholders

The following stakeholders are involved in the system:

1. **Customers:** End-users who browse and purchase products.
2. **Administrators:** Users who manage the platform, including product and order management.
3. **Developers:** The team responsible for building and maintaining the system.
4. **Delivery Personnel:** Individuals responsible for delivering products to customers (for future enhancements).

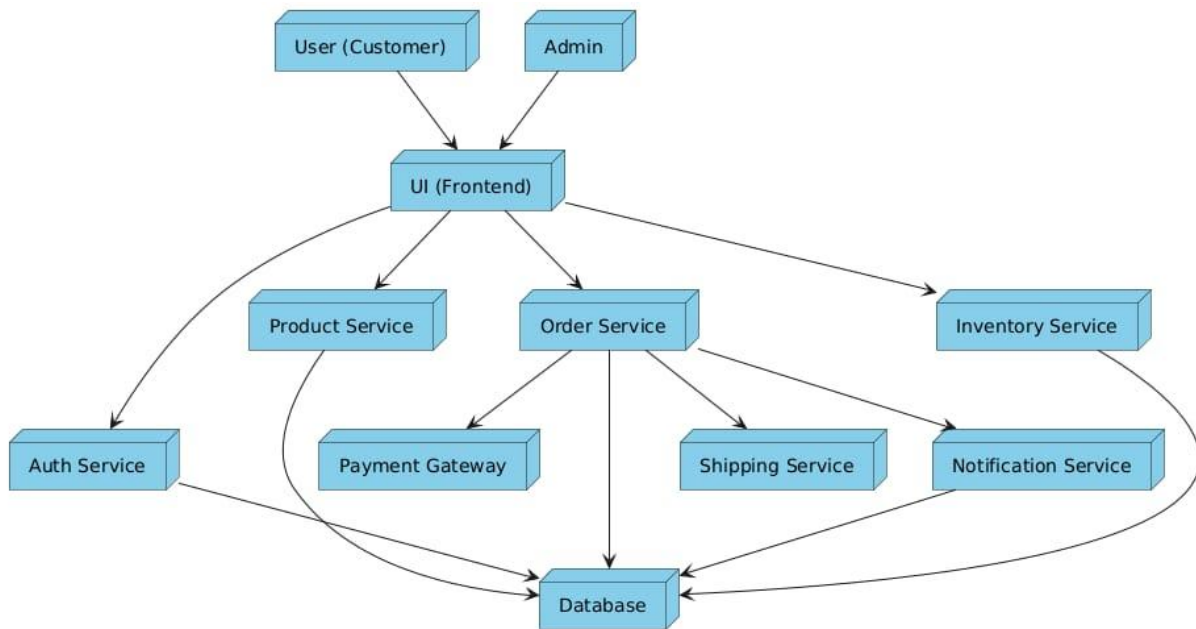
# System Design

It is a process of planning a new business system or replacing an existing system by defining its components or modules to satisfy the specific requirements. Before planning, we need to understand the old system thoroughly and determine how computers can best be used in order to operate efficiently.

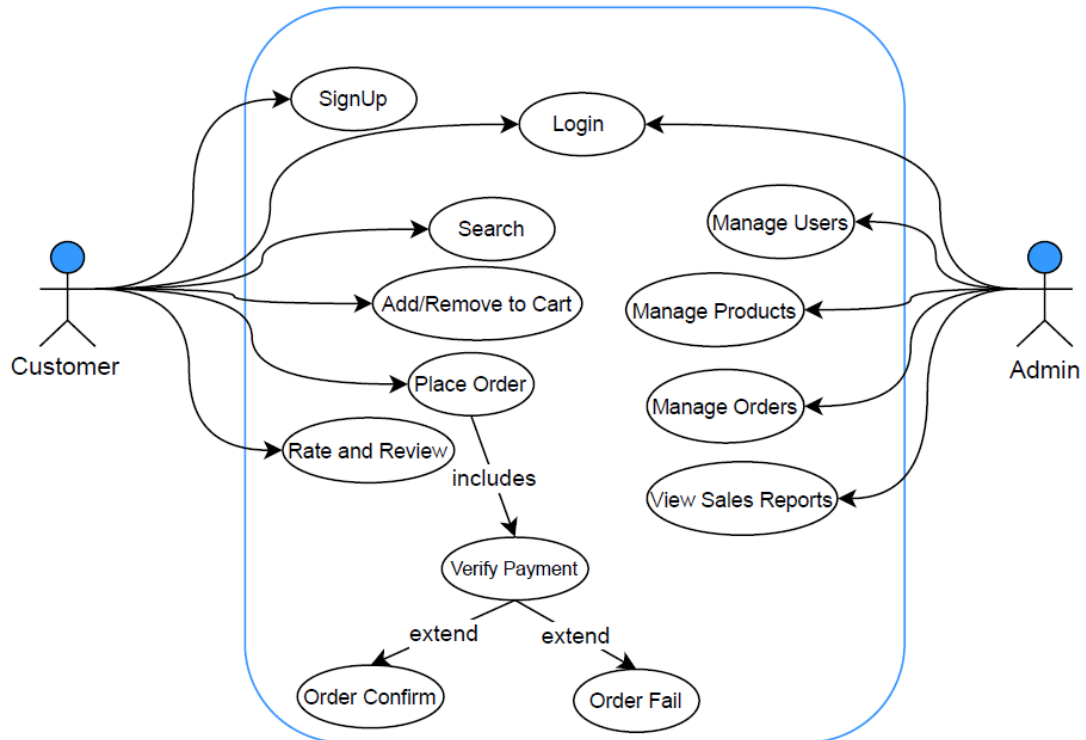
System Design focuses on how to accomplish the objective of the system. It mainly focuses on:

- ❖ Systems
- ❖ Processes
- ❖ Technology

## Block Diagram

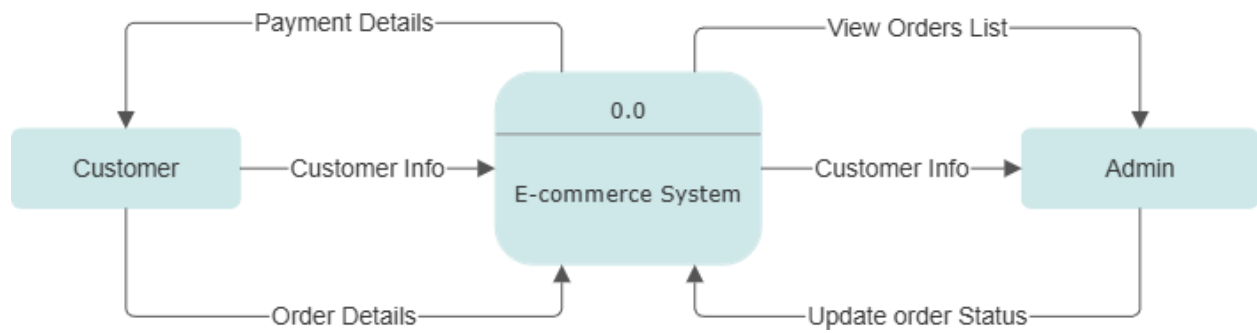


## Use Cases

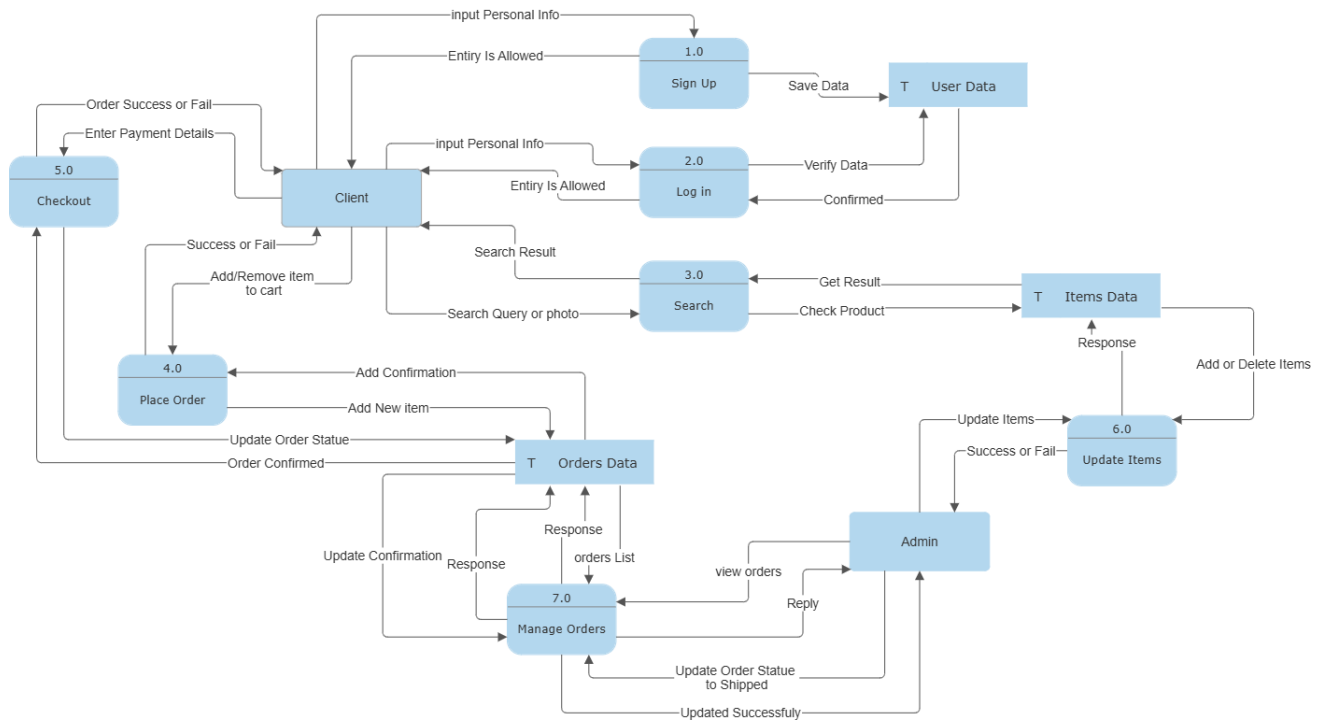


## DATA FLOW DIAGRAM (DFD)

DFD level 0 (Context Diagram):

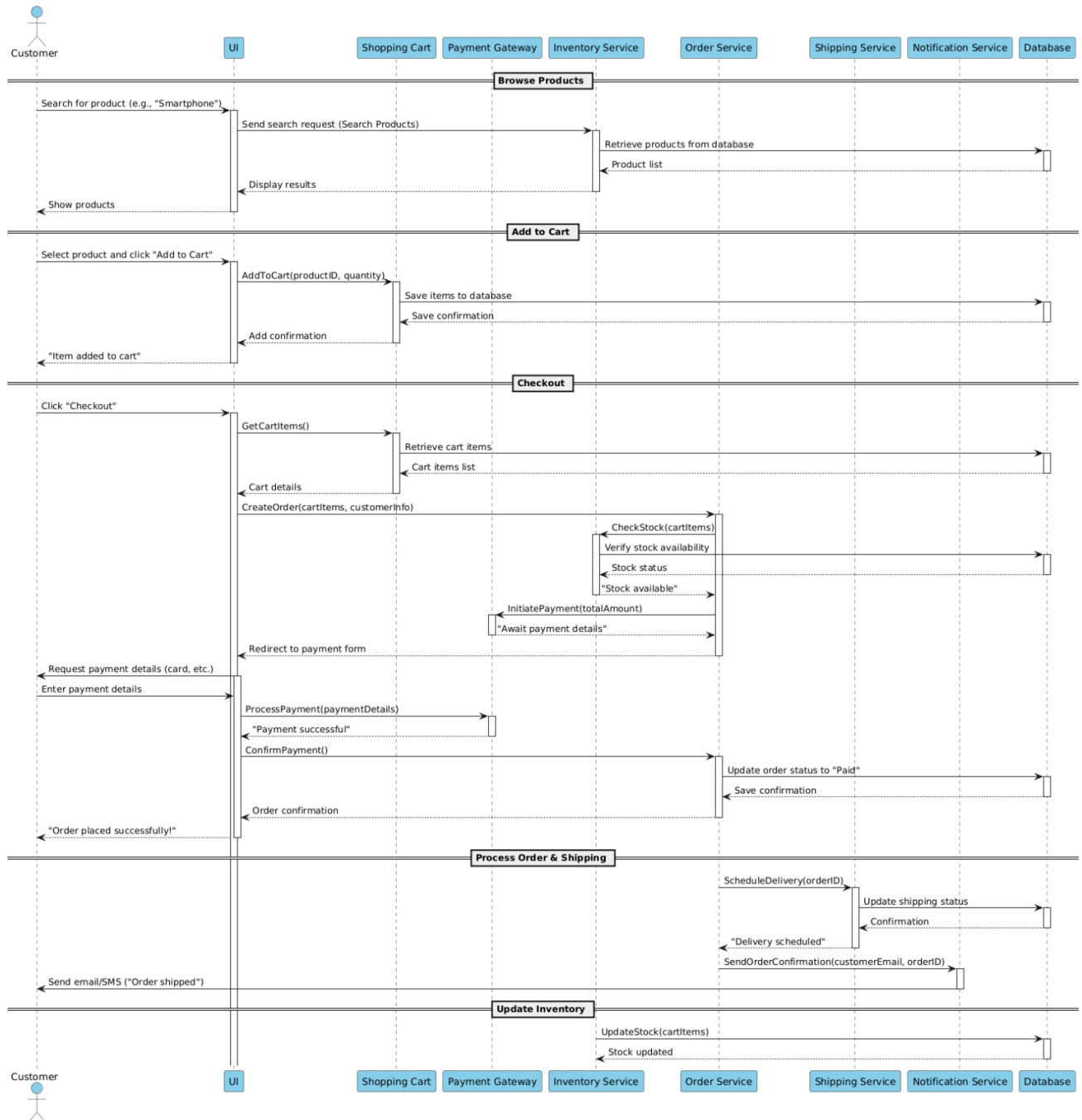


## DFD level 1:

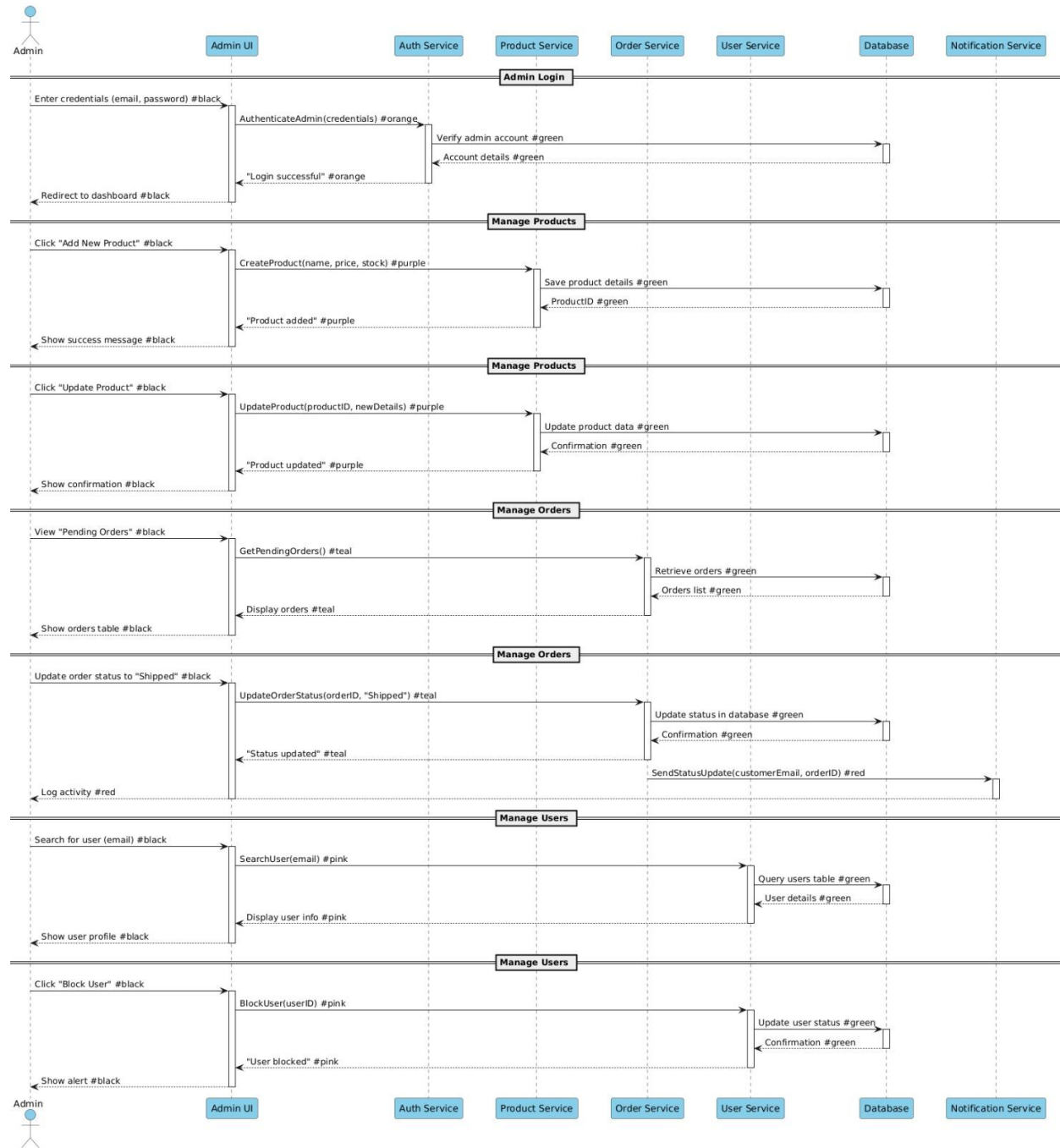


# Sequence Diagram

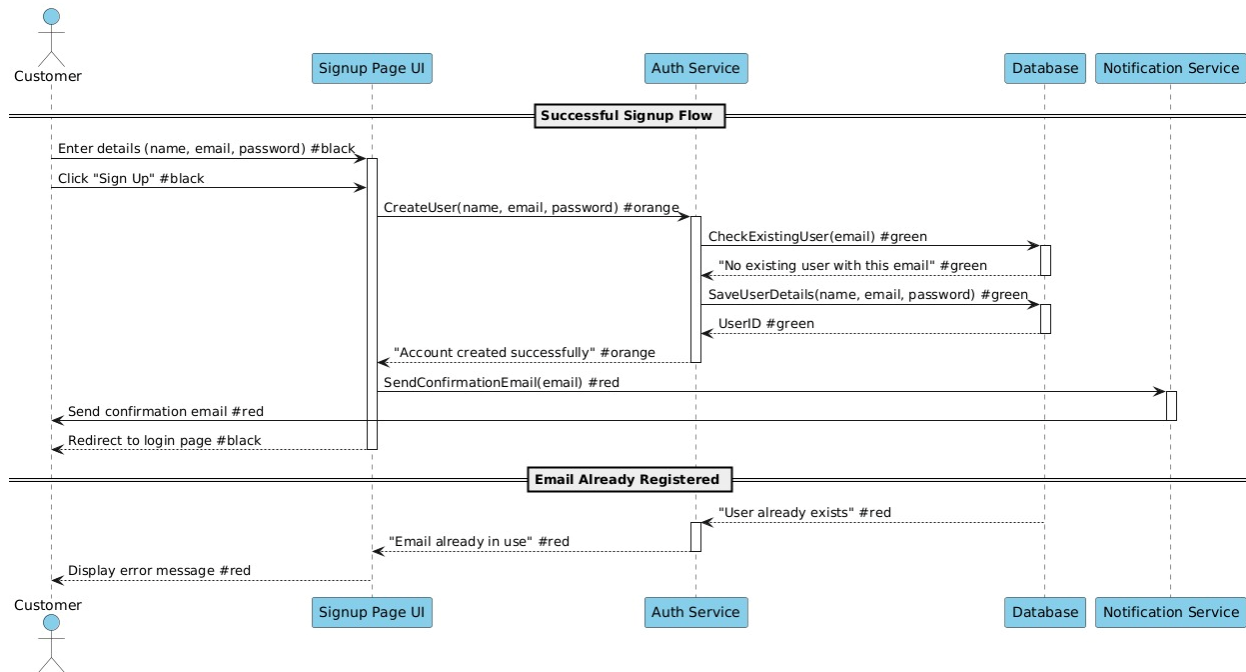
For customer:



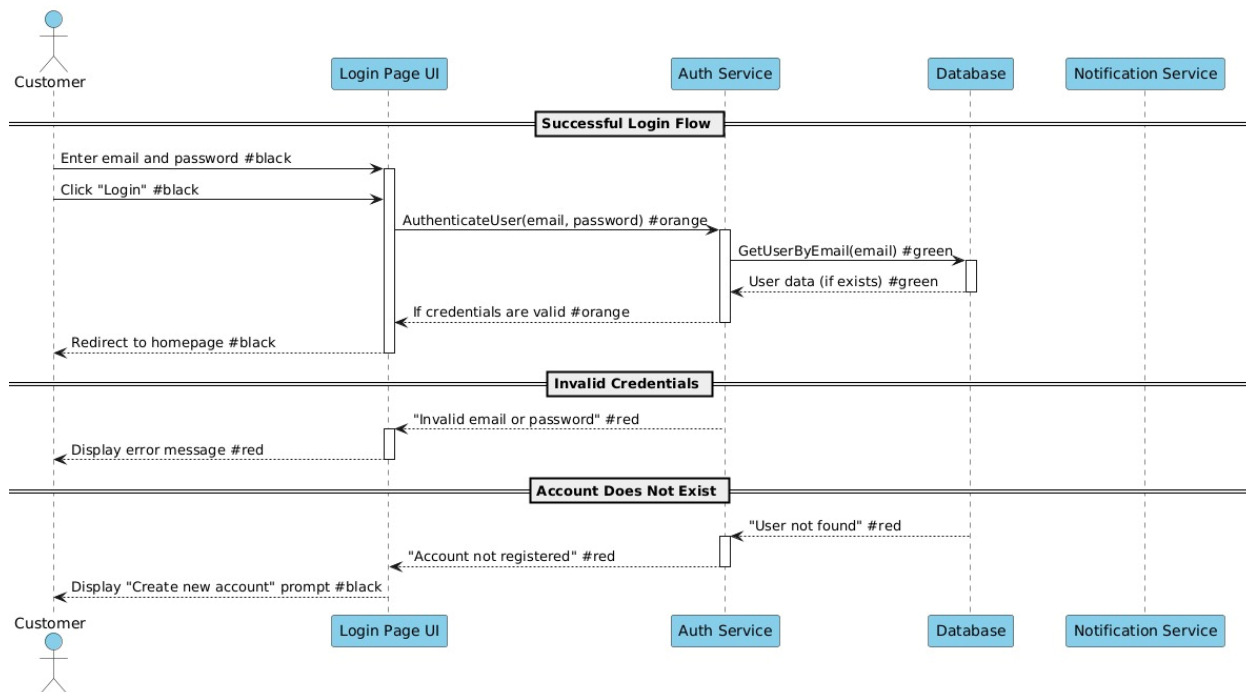
## For Admin:



## Signup Sequence:

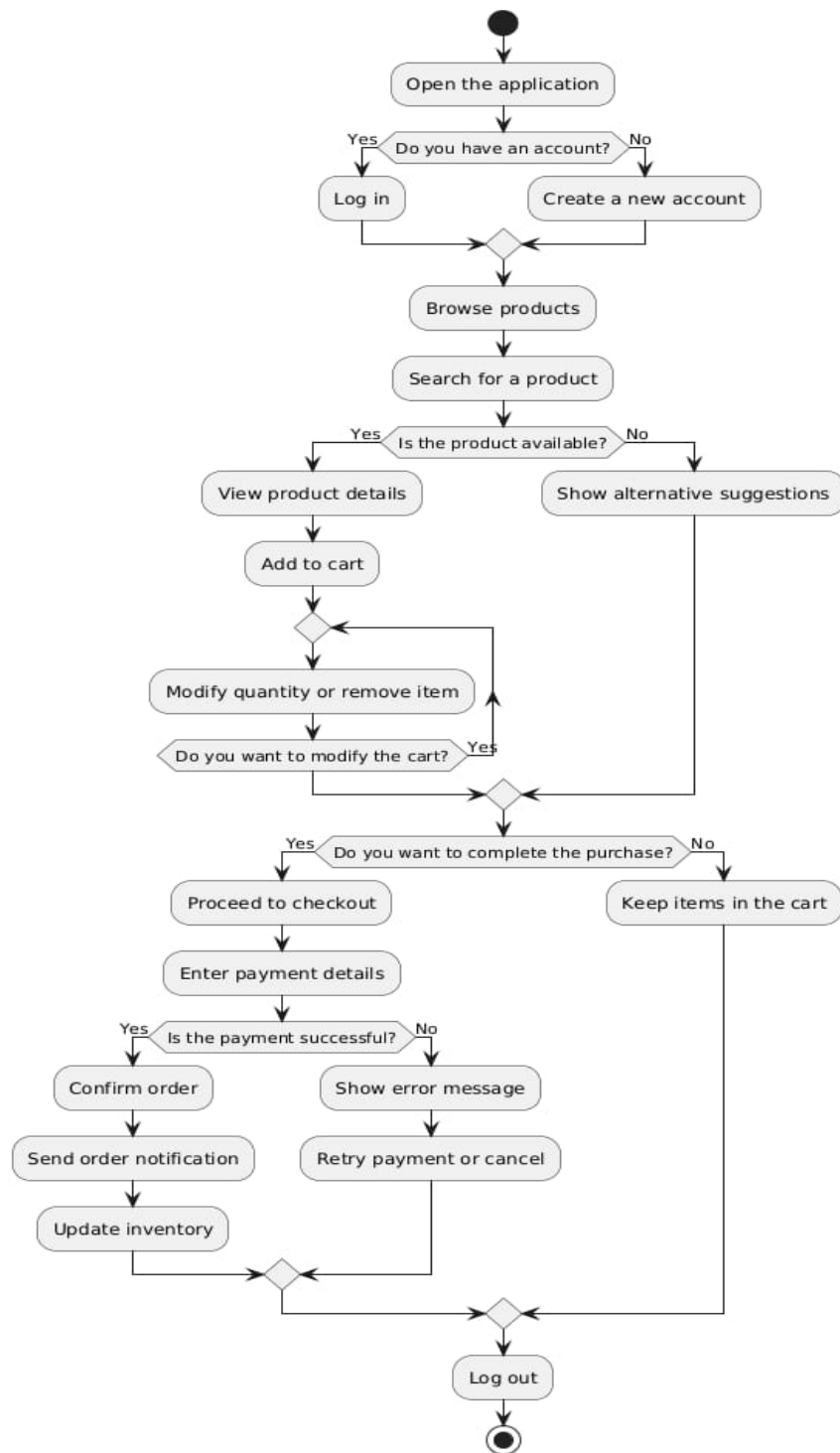


## Login Sequence:

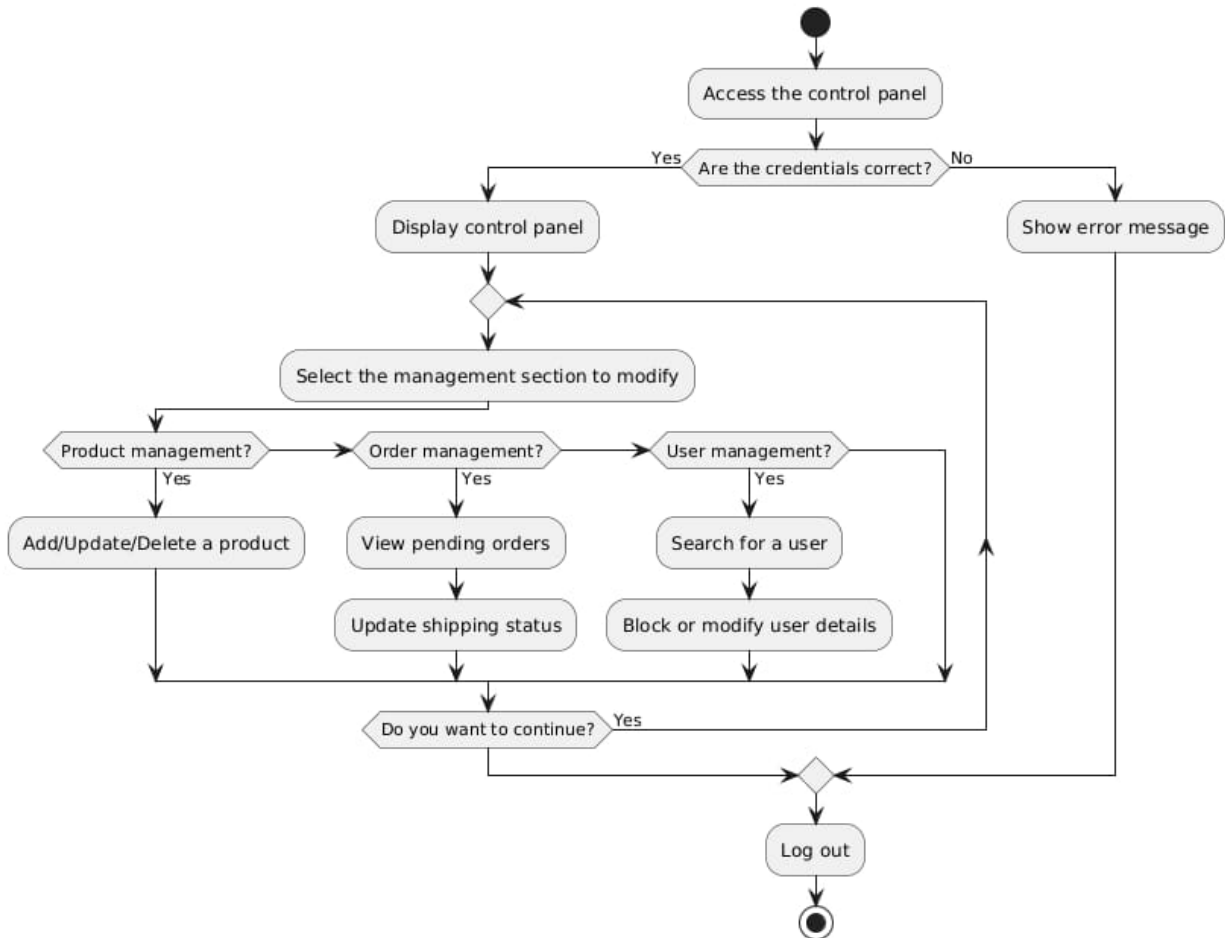


## Activity Diagram

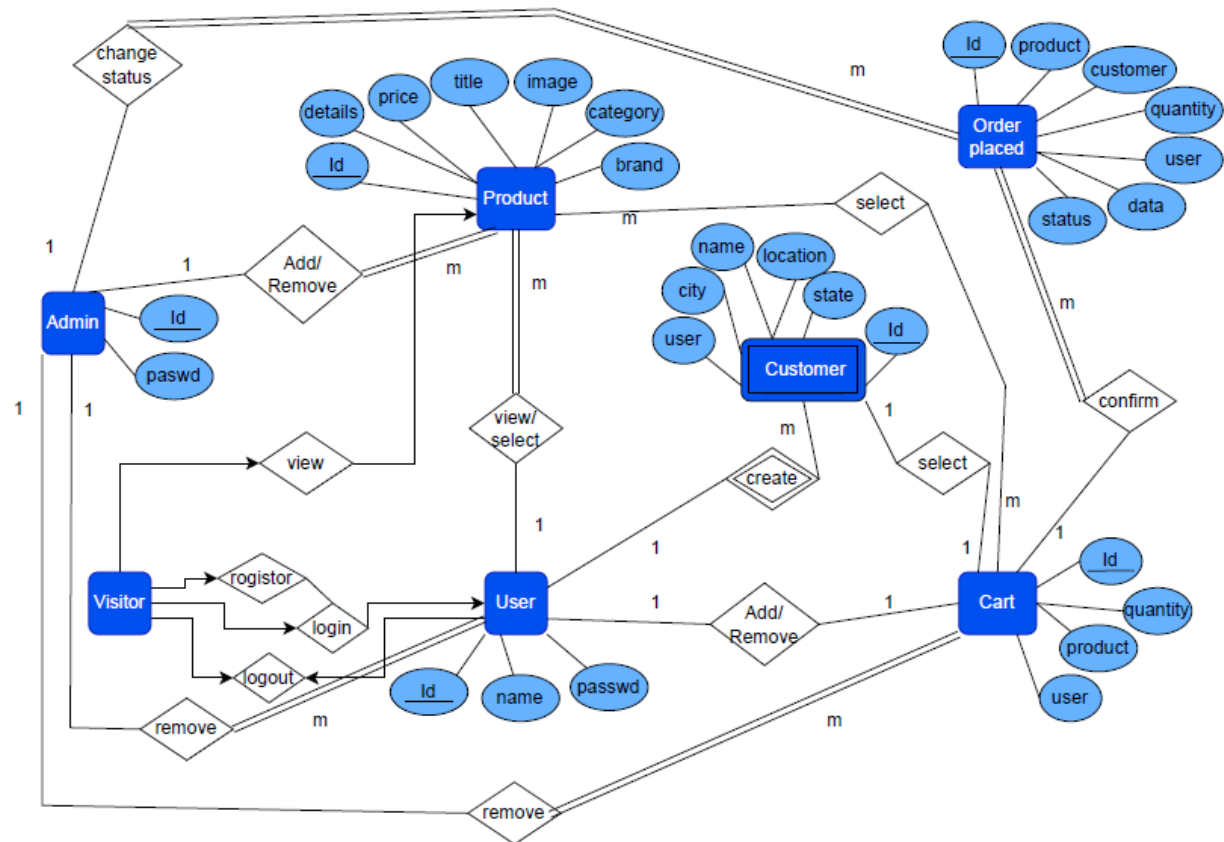
For customer:



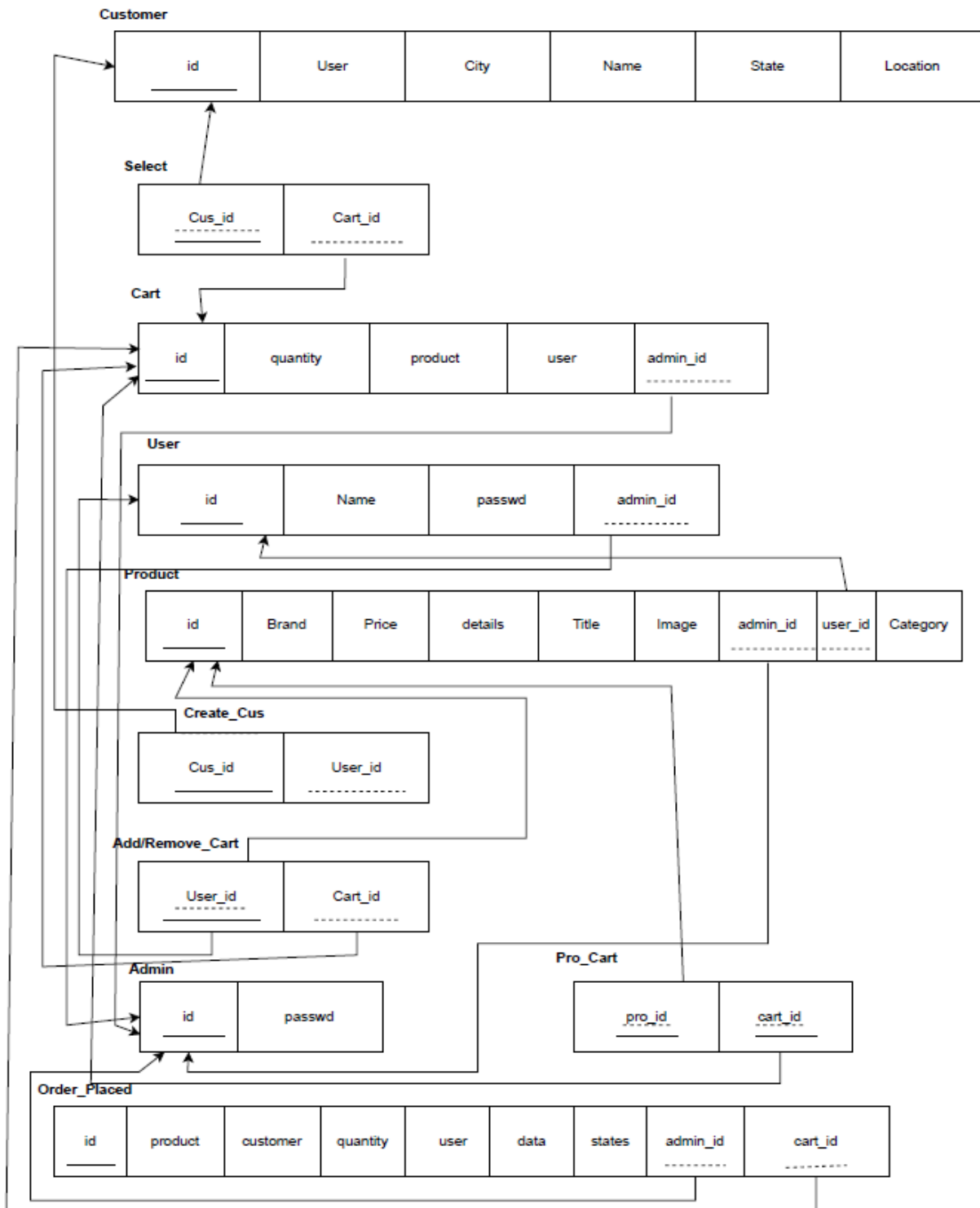
## For Admin:



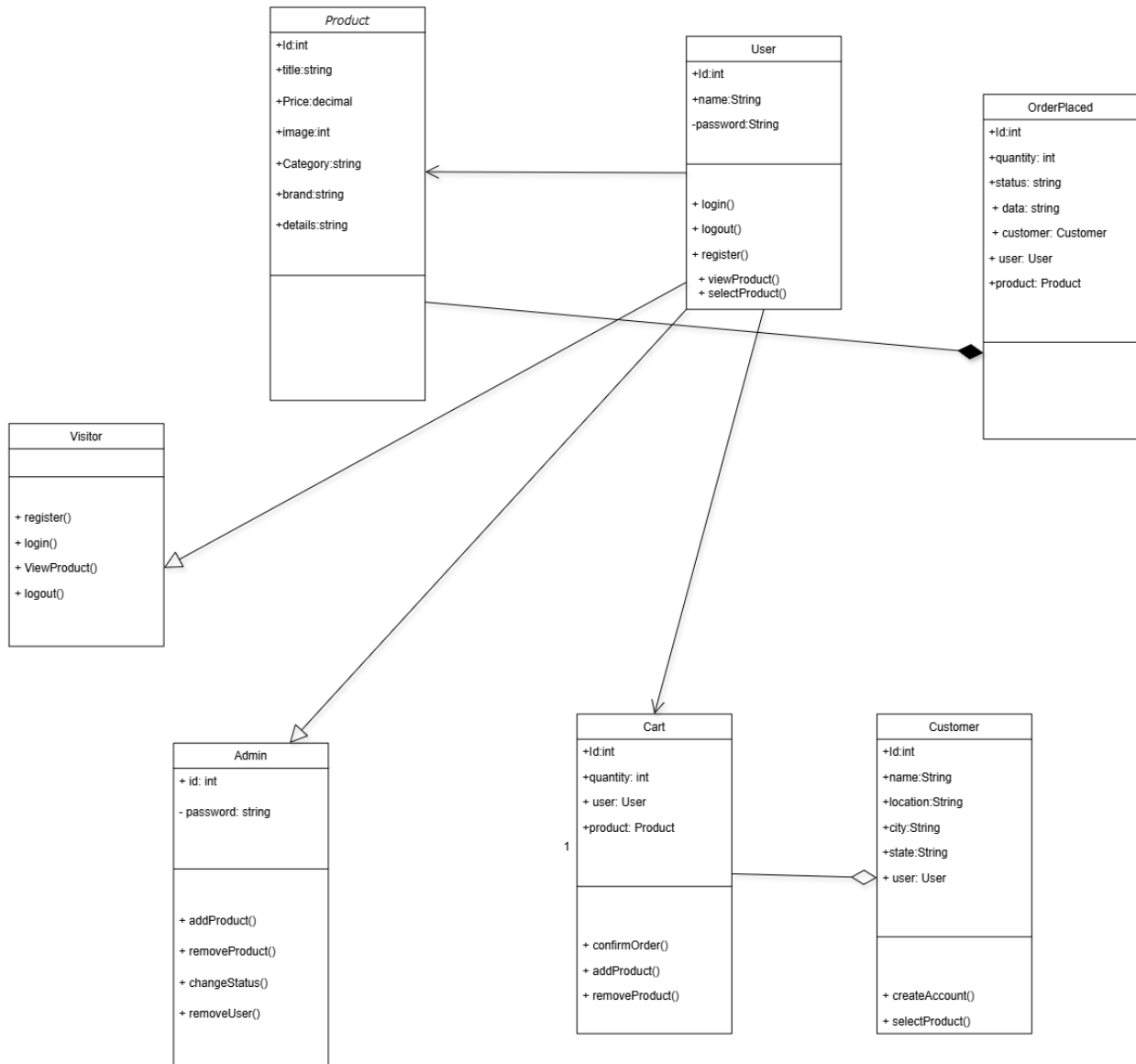
## Entity Relationship Model (ERD)



## Mapping Diagram



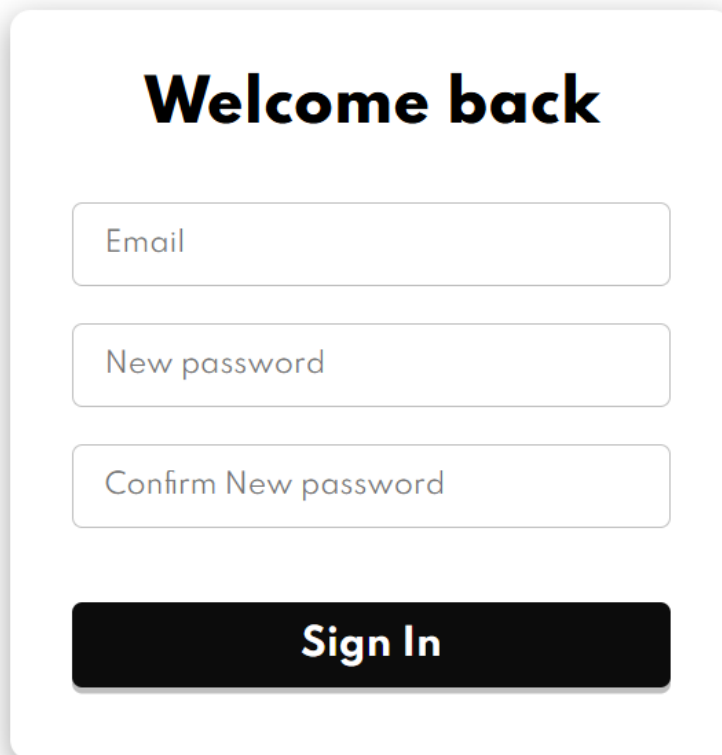
# Class Diagram



# CHAPTER THREE

# Website Design

## SIGNUP PAGE



**Welcome back**

Email

New password

Confirm New password

**Sign In**

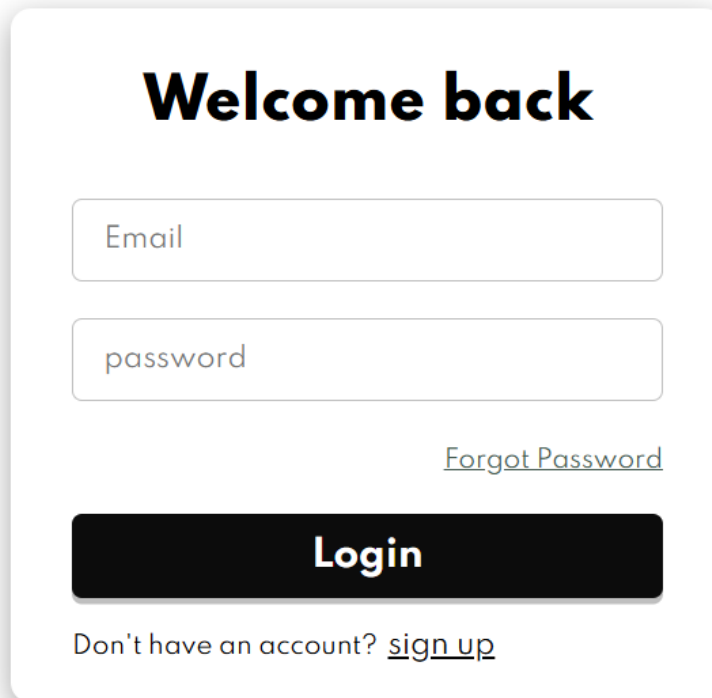
The **sign-up** page is designed to capture essential user information in a straightforward manner. It includes the following fields:

**Email:** A field is provided for users to enter their email address. This serves as a primary method of communication and account recovery.

**Password:** A secure password is required to protect the user's account. Passwords should adhere to security standards and provide robust protection.

**Confirm Password:** To ensure accuracy and prevent errors, users are asked to re-enter their chosen password in the confirm password field.

## LOGIN PAGE

A login page UI mockup featuring a white card with rounded corners and a subtle drop shadow. At the top, the text "Welcome back" is displayed in a bold, black, sans-serif font. Below this, there are two input fields: the first is labeled "Email" and the second is labeled "password", both in a light gray font. To the right of the password field, there is a link that says "Forgot Password" in a smaller, blue, underlined font. Below the input fields is a large, black, rounded rectangular button with the word "Login" in white, bold, sans-serif font. At the bottom of the card, there is a line of text that reads "Don't have an account? [sign up](#)", where "sign up" is a blue, underlined link.

**Welcome back**

Email

password

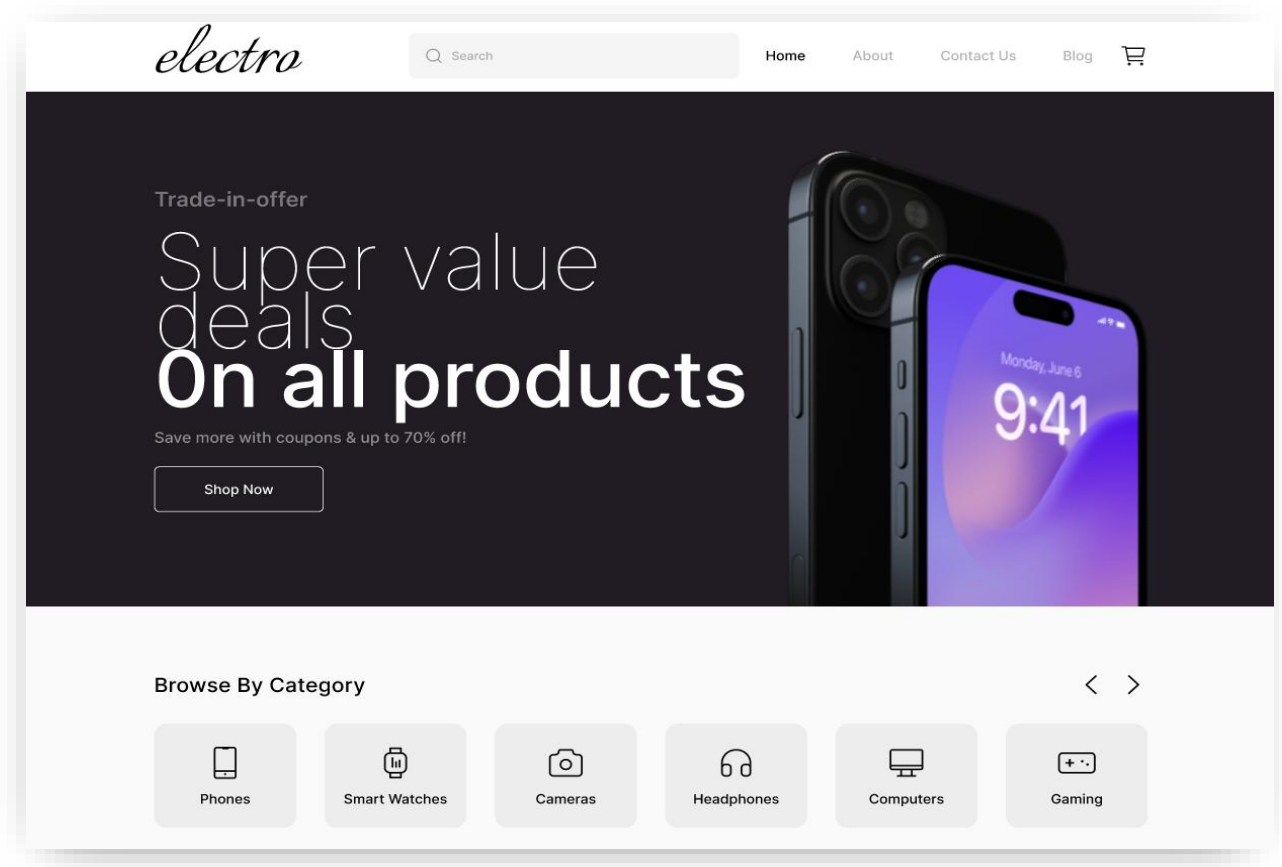
[Forgot Password](#)

**Login**

Don't have an account? [sign up](#)

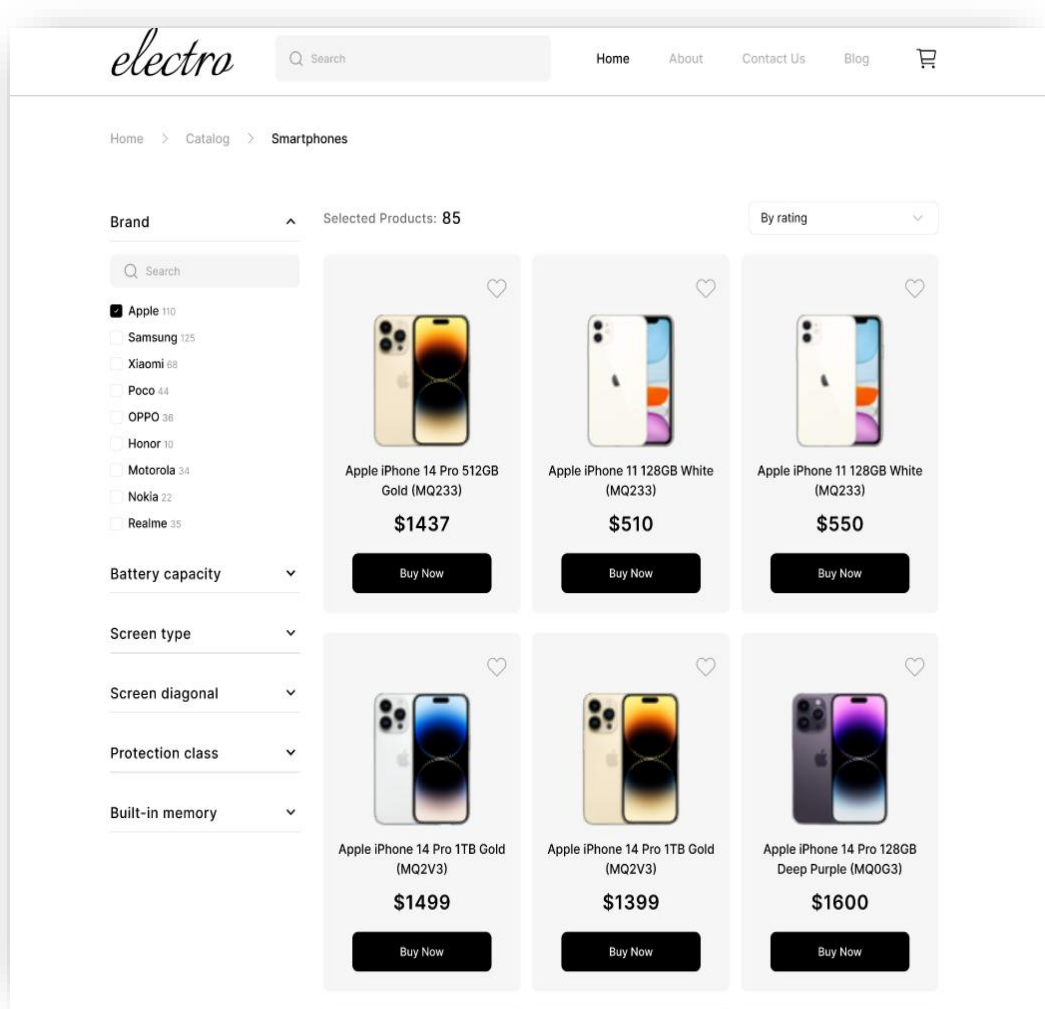
The **login** page is designed with simplicity and user-friendliness in mind. It features only two essential fields: a field for the user's username or email and a password field. Users are required to enter their registered username or email in the first field and provide their confidential password in the second field. Once the required information is entered, users can proceed to log in by clicking the "Login" button.

## Home Page



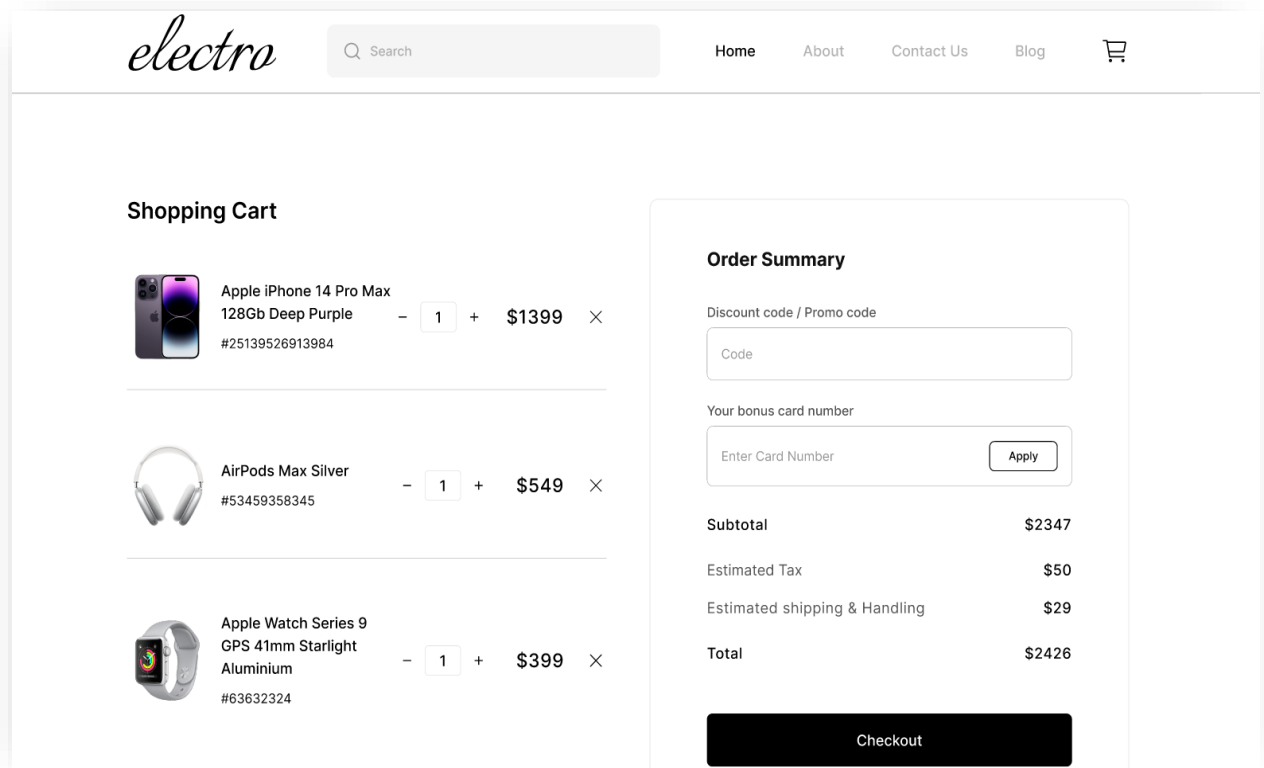
The home page of our e-commerce platform features a modern, user-friendly interface with a navigation bar for easy access to **Home**, **About**, **Contact Us**, **Blog**, and **shopping cart**. A prominent banner promotes "Super Value Deals on All Products" with discounts up to 70% off and a featured smartphone. Below, interactive icons categorize products into **Phones**, **Smart Watches**, **Cameras**, **Headphones**, **Computers**, and **Gaming**, ensuring seamless navigation and an intuitive shopping experience.

# PRODUCTS PAGE



The Products Page offers a structured layout for easy product browsing and filtering. It includes a **sidebar filter panel** where users can refine searches by **brand** (e.g., Apple, Samsung), **battery capacity**, **screen type**, **screen size**, **protection class**, and **built-in memory**. A search bar allows quick access to specific brands or features. This design enhances user experience by ensuring efficient navigation and seamless product discovery. Let me know if you'd like to add this to your document.

## Cart Page



### Cart Items Section:

- Displays selected products with images, names, specifications, and unique product IDs.
- Quantity adjustment available through "+" and "-" buttons.
- Remove (X) button for deleting items from the cart.
- Shows individual price for each product.

### Order Summary Section:

- Option to enter discount codes or bonus card numbers for offers.
- Displays:
  - **Subtotal:** Total before taxes and shipping.
  - **Estimated Tax:** Displays additional costs.
  - **Estimated Shipping & Handling:** Shows delivery charges.
  - **Total Amount:** Final payable amount after all charges.
- **Checkout Button:** Directs users to payment and final confirmation.

# Mobile Design

## SIGNUP PAGE

**Welcome back**

**Sign In**

## LOGIN PAGE

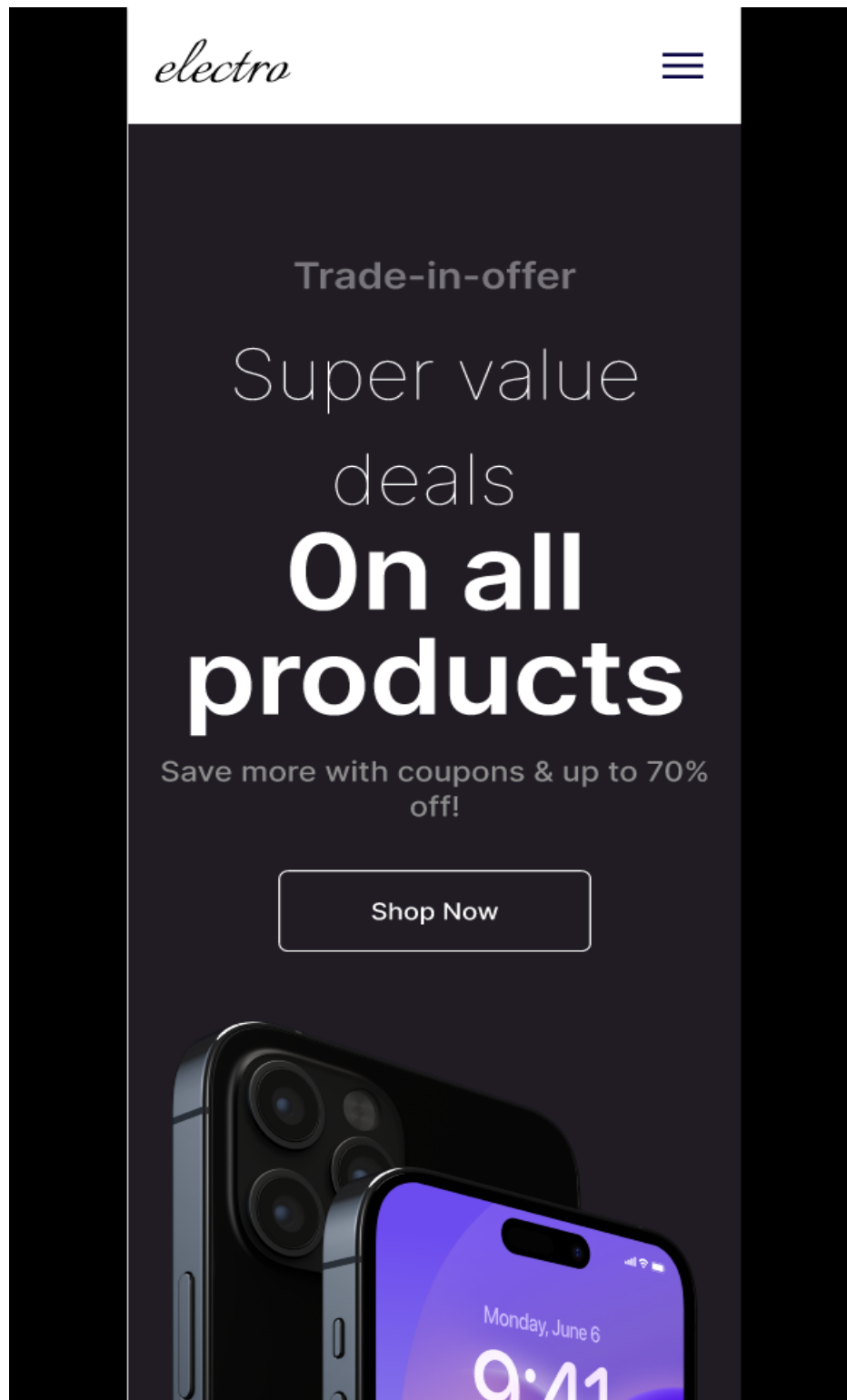
**Welcome back**

[Forgot Password](#)

**Login**

Don't have an account? [sign up](#)

# HOME PAGE



# Cart Page

*electro*



## Shopping Cart



Apple iPhone 14 Pro Max  
128Gb Deep Purple

#25139526913984

- 1 + \$1399 ×



AirPods Max Silver

#53459358345

- 1 + \$549 ×



Apple Watch Series 9 GPS  
41mm Starlight Aluminium

#63632324

- 1 + \$399 ×

## Order Summary

Discount code / Promo code

# Products Page

*electro*

Filters

By rating

Products Result : 85



Apple iPhone 14 Pro 512GB Gold...

**\$1437**

Buy Now



Apple iPhone 11 128GB White (M...

**\$510**

Buy Now



Apple iPhone 11 128GB White...

**\$550**

Buy Now



Apple iPhone 14 Pro 1TB Gold (M...

**\$1499**

Buy Now

# Payment Page

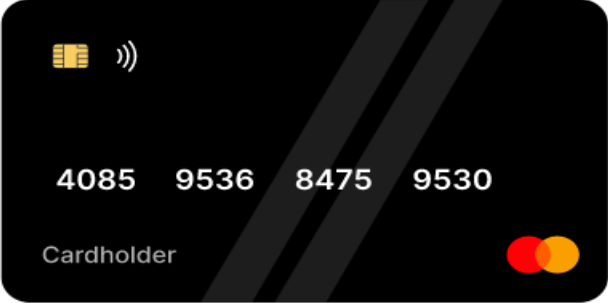
*electro*

 Step 2  
Shipping

 Step 3  
Payment

## Payment

Credit CardPayPalPayPal Credit



4085 9536 8475 9530

Cardholder

☒ Same as billing address

Back

Pay

# CHAPTER FOUR

## Introduction

This chapter presents a summary of the key findings from the e-commerce project focused on selling electronics. It discusses the significance of the project, its practical applications, limitations, and suggestions for future improvements.

## Main Findings

The project successfully developed an e-commerce platform with both Android and web applications for the Middle Eastern market. The integration of **Kotlin** for the Android app, **react** for the web frontend, and a **.NET backend** ensured smooth and efficient user experience. Key functionalities such as product browsing, adding items to the cart, checkout, and order tracking were implemented with an emphasis on performance, scalability, and security.

## Why is this Project Important

This project addresses the growing demand for online electronics shopping in the Middle East. It offers a convenient, user-friendly, and efficient solution for consumers seeking electronic products. By utilizing modern technologies like **Kotlin**, **React**, and **.NET**, the project is positioned to compete with established e-commerce platforms, providing users with a robust alternative for their shopping needs.

# Testing

The testing process ensures the e-commerce platform functions as expected and maintains high performance across both web and Android applications. A combination of various testing methods was implemented to verify the system's reliability, performance, and security. The main testing approaches include:

## Unit Testing

Unit testing was carried out to verify the functionality of individual components within the system. Each feature, such as the product listing display, cart management, and payment modules, was tested in isolation to ensure they performed correctly. Unit tests were automated to speed up the development process and to identify issues in early stages, ensuring a smooth user experience.

## Integration Testing

Integration testing focused on verifying the seamless interaction between the front-end and back-end systems. Specifically, communication between the React-based web frontend, the Kotlin-based Android app, and the .NET backend APIs was tested. This approach ensured that user actions such as adding items to the cart, submitting orders, and processing payments were correctly communicated between all layers of the system.

## Regression Testing

To ensure that updates or changes did not introduce new issues, regression testing was conducted. After each code update or feature addition, the entire system was re-tested to check for any disruptions in previously working functionalities. This process helped in identifying recurring issues and addressing them promptly, reducing the risk of bugs after deployments.

## Performance Testing

Performance testing was done to evaluate the platform's ability to handle high traffic loads and ensure its responsiveness. Key areas of focus included system speed, load handling, and overall user experience. This testing also included stress testing to simulate extreme conditions, ensuring the system could function well even under peak user load.

## Security Testing

Security testing was essential to protect sensitive user data and ensure secure transactions. This included validating the implementation of encryption techniques for user data, secure login protocols, and the integrity of payment systems. Testing for vulnerabilities like SQL injection, cross-site scripting (XSS), and other security risks were performed to safeguard both user data and business transactions.

# Backend (testing endpoints)

The image shows the Swagger UI interface for the Store.G06.APIs v1 definition. The browser address bar displays `https://localhost:7016/swagger/index.html`. The Swagger logo is in the top left, and a dropdown menu in the top right shows "Select a definition" with "Store.G06.APIs v1" selected. The main heading is "Store.G06.APIs 1.0 OAS3" with the URL `https://localhost:7016/swagger/v1/swagger.json` below it. The interface is divided into two sections: "Accounts" and "Basket".

**Accounts**

- POST `/api/Accounts/login`
- POST `/api/Accounts/register`
- GET `/api/Accounts/GetCurrentUser`
- GET `/api/Accounts/Address`
- PUT `/api/Accounts/Address`

**Basket**

- GET `/api/Basket/{id}`
- POST `/api/Basket`

An "Activate Windows" watermark is visible in the bottom right corner of the Swagger UI area.

The image shows the Swagger UI interface for the `GET /api/Products` endpoint. The browser address bar displays `https://localhost:7016/swagger/index.html`. The endpoint is selected, and the "Parameters" tab is active. A "Cancel" button is in the top right corner of the parameters section.

| Name                                     | Description |
|------------------------------------------|-------------|
| Search<br>string<br>(query)              | Search      |
| Sort<br>string<br>(query)                | Sort        |
| BrandId<br>integer(\$int32)<br>(query)   | BrandId     |
| TypeId<br>integer(\$int32)<br>(query)    | TypeId      |
| PageSize<br>integer(\$int32)<br>(query)  | PageSize    |
| PageIndex<br>integer(\$int32)<br>(query) | PageIndex   |

At the bottom of the parameters section, there is an "Execute" button and a "Clear" button. An "Activate Windows" watermark is visible in the bottom right corner of the Swagger UI area.

# Practical Implementations

The platform is designed with the following practical implementations:

## User Interface (UI):

Clean and intuitive UI on both the Android and web platforms for easy navigation. The design focuses on simplicity and accessibility, ensuring a user-friendly experience across devices.

## Product Catalog:

A dynamic catalog to display various electronics with advanced filtering and sorting capabilities. Users can browse through categories, brands, prices, and specifications, making it easier to find the desired product.

## Payment Integration:

Secure payment gateways for a smooth transaction experience. The system ensures that user payment information is securely processed with encryption and comply with industry standards.

## Order Tracking:

Real-time order tracking to provide users with updates on their purchases. Notifications are sent to users at key stages of their order journey, such as when the order is confirmed, shipped, and delivered.

**Backend Architecture:** A robust backend built using .NET to handle user requests, data storage, and interactions between the frontend and database. The backend is designed to scale efficiently as the user base grows.

**AI-Powered Product Recommendations:** An AI-driven recommendation engine that analyzes user behavior and preferences to suggest relevant products. By using machine learning algorithms, the system can personalize product suggestions, boosting user engagement and increasing sales. This feature adapts to the user's shopping history and search patterns to recommend products they are more likely to purchase.

## Limitations

Despite its success, the project has certain limitations:

- **Regional Constraints:** The platform currently caters only to the Middle East, limiting its market reach.
- **Payment Gateway Restrictions:** The payment system is limited to certain countries in the Middle East, which may affect user accessibility.
- **Scalability:** Although the system is designed to handle a moderate load, scaling to a larger audience will require additional infrastructure and optimization.

## Future Recommendations

To improve the platform and expand its user base, the following recommendations are made:

- **Global Expansion:** The platform could be expanded to other regions with localized payment systems and shipping options.
- **Mobile App Features:** Enhancements such as push notifications for promotions, personalized recommendations, and social media integrations could be added.
- **Backend Optimization:** Future updates should focus on optimizing the backend for faster response times and increased scalability.
- **AI/ML Integration:** Incorporating AI or machine learning could enhance product recommendations and improve user experience by personalizing the shopping experience.

## Conclusion Summary

In conclusion, this project successfully created an e-commerce platform that meets the needs of electronic shoppers in the Middle East. The combination of modern technologies, including Kotlin for the Android app, react for the web frontend, and **.NET** for the backend, alongside a strong focus on user experience, has resulted in a robust platform.

One of the standout features is the **AI-powered product recommendation engine**, which personalizes the shopping experience for users by analyzing their behavior and preferences. This enhances the platform's ability to provide relevant product suggestions, leading to improved user engagement and higher conversion rates.

With further improvements, such as expanding to new regions, enhancing the AI capabilities, and implementing more advanced features like personalized promotions and better scalability, the project holds great potential for future growth and success.